

DIMENSION STRATIFICATION OF κ_{ch} ON THE CALABI–YAU LANDSCAPE

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ABSTRACT. SCAFFOLD STUB. For compact Calabi–Yau d -folds, the chiral-shadow invariant $\kappa_{\text{ch}}(\mathcal{A}_X)$ equals the Hodge supertrace $\sum_q (-1)^q h^{0,q}(X)$ (thm:kappa-hodge-supertrace-identification) unconditionally via HKR plus the Mukai pairing plus an HC_d^- trace. The dimension stratification $d \in \{1, 2, 3, 4, 5\}$ then reads off explicitly (thm:kappa-stratification-by-d): elliptic curves contribute 0; K_3 contributes 2; abelian and bielliptic contribute 0; quintic, $K_3 \times E$, $E^{\times 3}$ contribute 0; local $\mathbb{P}^2$ contributes $3/2$ (thm:local-p2-shadow); CY_4 sextic contributes 2; CY_5 generic 0. The conifold is *not* a local surface (it has resolved form $\text{Tot}(\mathcal{O}(-1)^2 \rightarrow \mathbb{P}^1)$, outside the $\text{Tot}(K_S \rightarrow S)$ family); the direct McKay route gives $\kappa_{\text{ch}}(\text{conifold}) = 1$.

PRIMARY THEOREMS

- thm:kappa-hodge-supertrace-identification (unconditional, compact CY, via HKR + Mukai + HC_d^-).
- thm:kappa-stratification-by-d (explicit tabulation $d = 1..5$).
- thm:local-p2-shadow $\kappa_{\text{ch}} = 3/2$.
- cor:conifold-non-local-surface (resolves the conifold-vs-local-surface classification).
- thm:borcherds-weight-kappa-BKM-universal ($\kappa_{\text{BKM}}(\Phi_N) = c_N(o)/2$ universal across $N \in \{1, 2, 3, 4, 6\}$; $N = 1$ coincidence with $\kappa_{\text{ch}} + \chi(\mathcal{O}_{\text{fiber}})$ fails for $N \geq 2$, resolving the naive $N = 1$ identification).

CHAPTER INPUTS (POINTERS)

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\input{../chapters/examples/cy_d_kappa_stratification.tex}
\input{../chapters/examples/derived_categories_cy.tex}
\input{../chapters/examples/matrix_factorizations.tex}
\input{../chapters/examples/k3e_cy3_programme.tex}
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CROSS-VOLUME DEPENDENCIES

- Vol I landscape_census.tex (canonical κ_{ch} formulas).
- Vol II \ref{part:ht-landscape} (HT landscape; compare κ_{ch} on HT-gauge side).